Tech Interview Prep:

Solving the "Lost in the Middle" Phenomenon & Advanced Re-Ranking in High-Stakes RAG

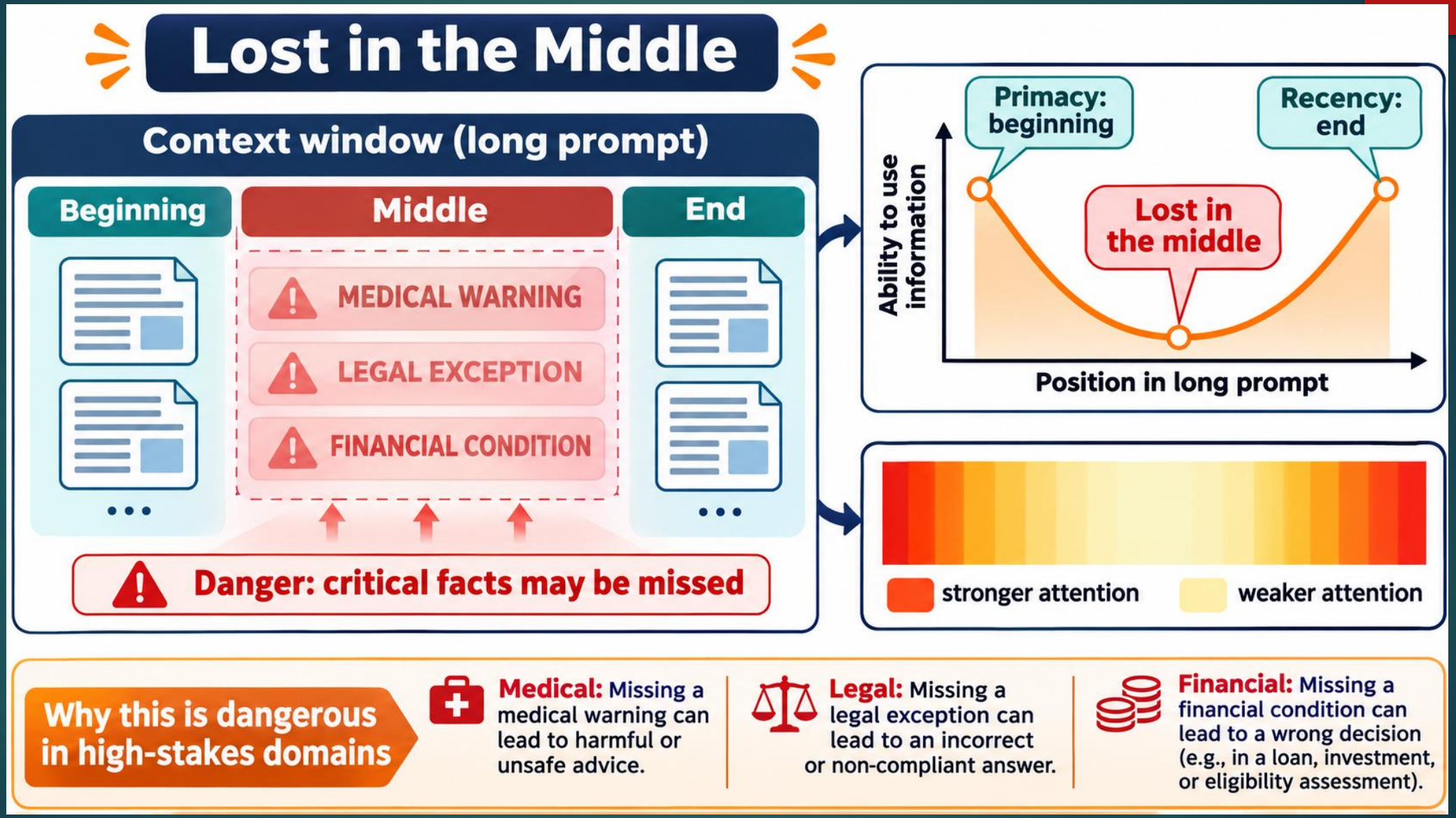
"Walk me through how you solve the 'Lost in the Middle' problem and how re-ranking improves retrieval precision in high-stakes domain applications."

Problems:
1) The search finds related but wrong information
2) The AI forgets information placed in the middle of a long prompt

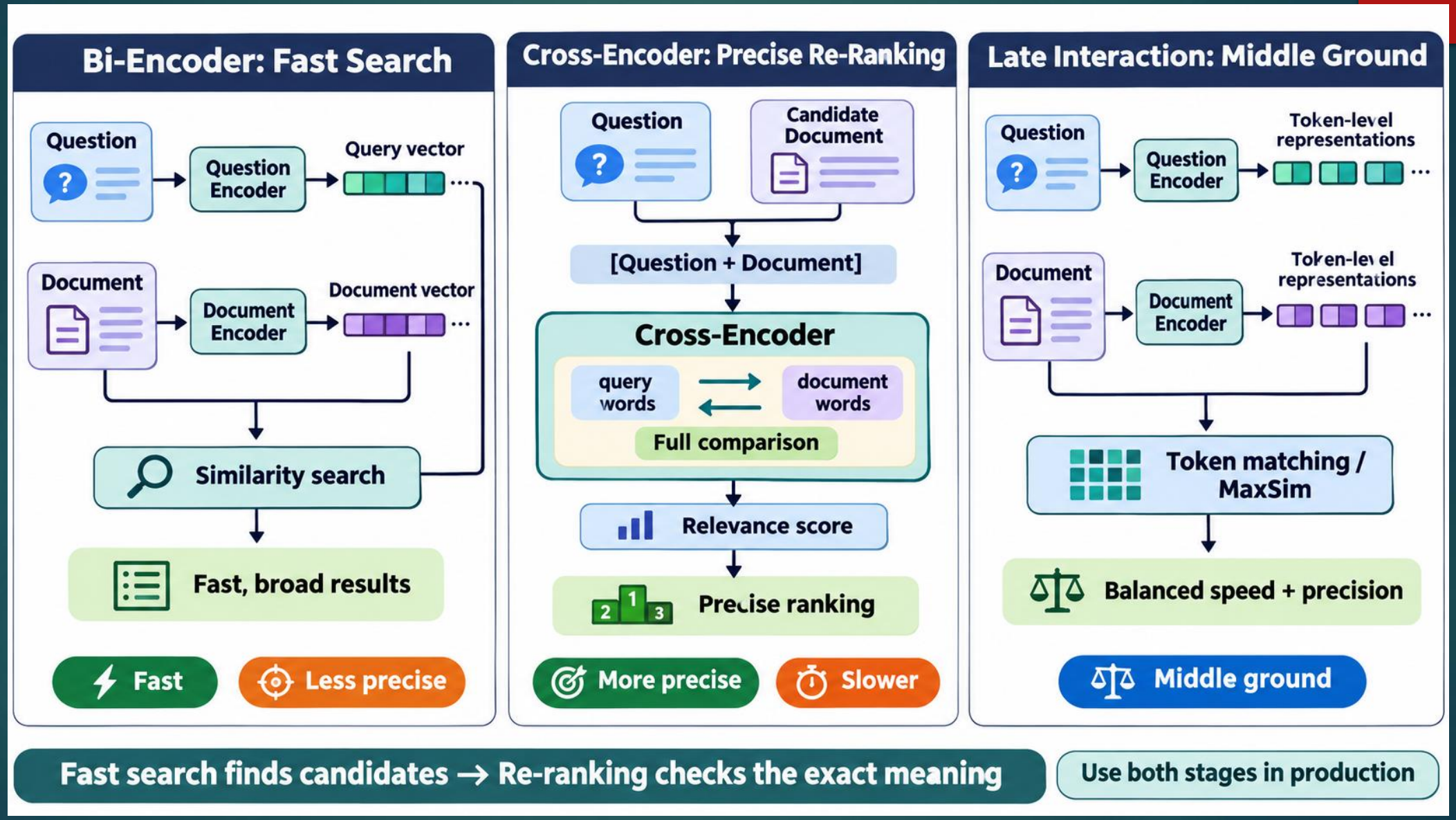
NAME: DARA SEN

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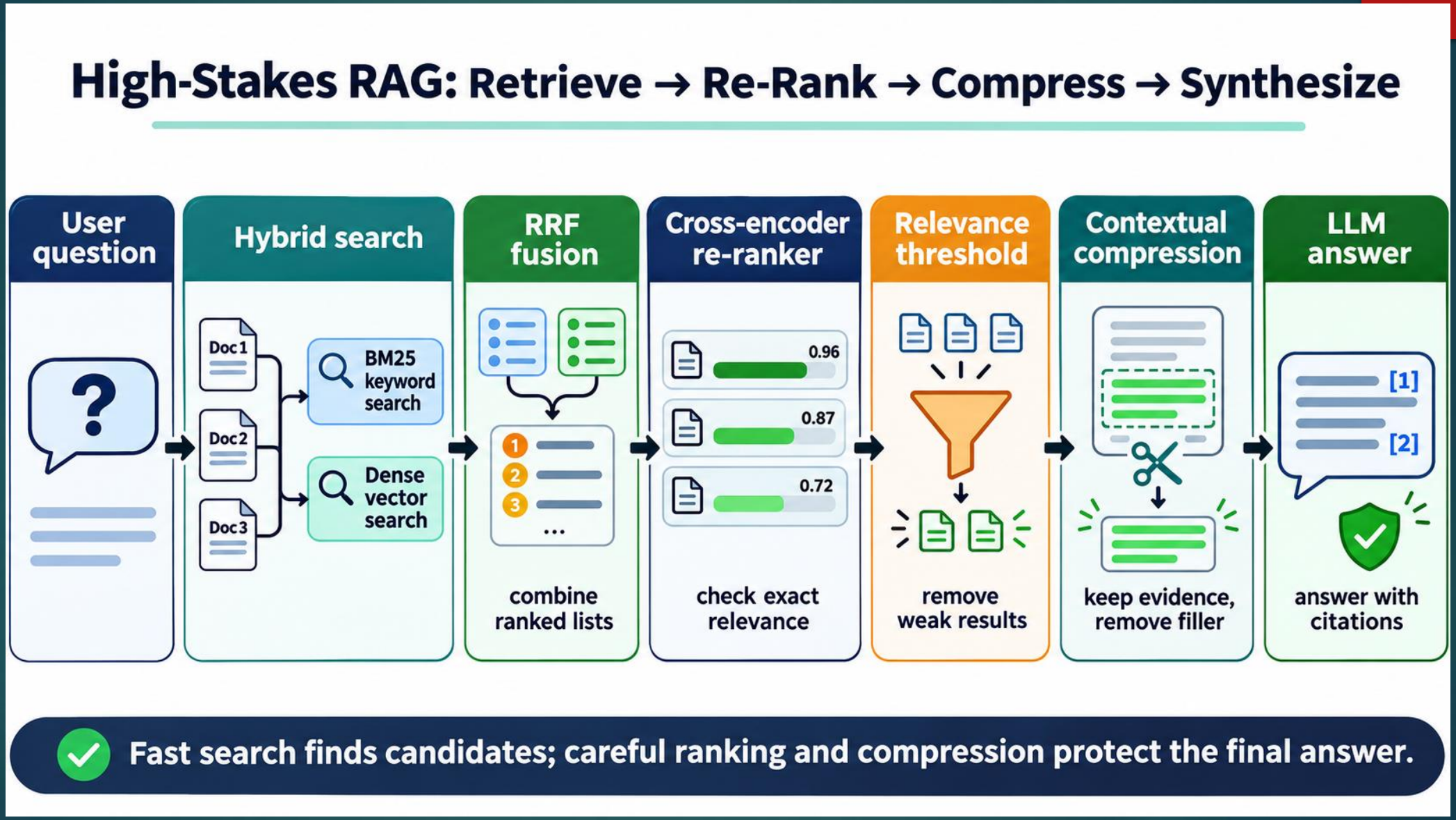
Deconstructing “Lost in the Middle”: Attention Mechanics



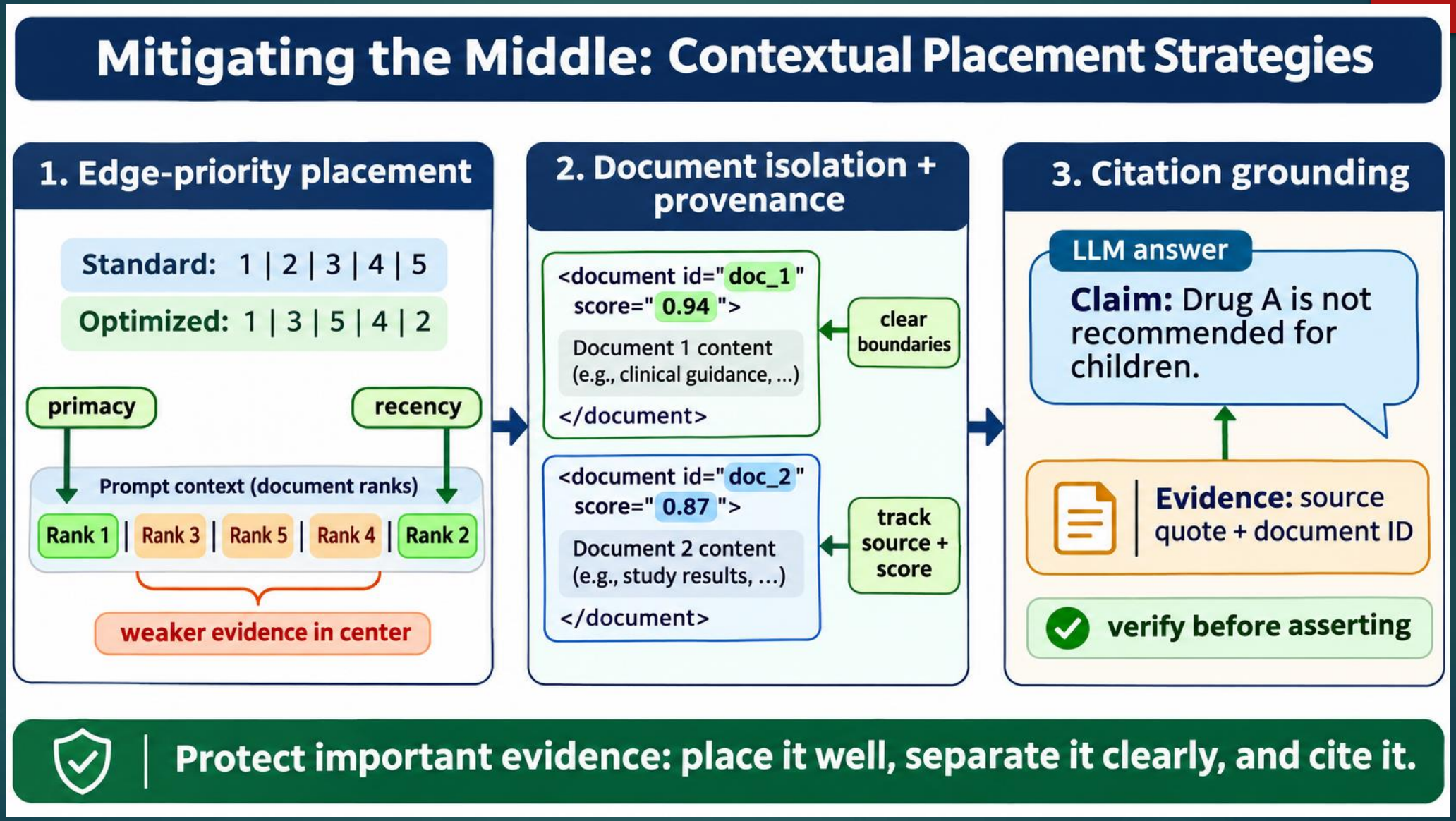
Two-Stage Retrieval: Bi-Encoders vs. Cross-Encoders



Production Pipeline for High-Stakes Domains



Mitigating the Middle: Contextual Placement Strategies



Summary & Enterprise Systems Takeaways

- Bi-encoder vs. Cross-encoder (Late interaction: keeping all token vectors)
- Lost in the Middle --> Edge-priority placement
- Citation grounding